

PAPER_1585 — Earth Core Radius = $R_{\text{core}} = A_5 \cdot SO_5 \cdot D_{\text{BSFG}} - SO_5^2 - D_{\text{BSFG}} - N_{\text{CH}} = 3600 - 100 - 6 - 9 = 3485$ km

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Geophysics **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209CC_S604 (Geophysics Unified Proof Set) lists **Earth Core Radius** as an EXACT-tier UQFF closure:

$$R_{\text{core}} = A_5 \cdot SO_5 \cdot D_{\text{BSFG}} - SO_5^2 - D_{\text{BSFG}} - N_{\text{CH}} = 3600 - 100 - 6 - 9 = 3485 \text{ km}$$

UQFF Closed Identity

$R_{\text{core}} = A_5 \cdot SO_5 \cdot D_{\text{BSFG}} - SO_5^2 - D_{\text{BSFG}} - N_{\text{CH}} = 3600 - 100 - 6 - 9 = 3485 \text{ km}$ EXACT
Pure integer-primitive expression with zero fitted constants.

Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.

Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER_1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF's predictive economy.

NOT REPLACEMENT

Empirical geophysics measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

Reference

- Source: PAPER_1209CC_S604
- Related: PAPER_1494-1575 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "earth_core_radius_3485_km"})`

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